D.K.T.E. SOCIETY'S TEXTILE AND ENGINEERING INSTITUTE, ICHALKARANJI

(An Empowered Autonomous Institute, Affiliated to Shivaji University, Kolhapur)

DEPARTMENT OF MCA

A PROJECT REPORT ENTITLED

ON

[PROJECT TITLE]

SUBMITTED IN PARTIAL FULFILLMENT OF THE REQUIREMENTS FOR THE DEGREE OF MASTER OF COMPUTER APPLICATION

BY

[Mr./Ms. Student Name]

[Mr./Ms. Student Name]

UNDER THE GUIDANCE OF

Mr./Ms. [Guide Name]

DKTE

Promoting Excellence in Teaching, Learning & Research

THROUGH THE DIRECTOR

D.K.T.E. S's TEXTILE AND ENGINEERING INSTITUTE, ICHALKARANJI – 416115

ACADEMIC YEAR: 2026–2027

CERTIFICATE

This is to certify that the project report entitled "[**PROJECT TITLE**]" is a record of the project work carried out by Mr./Ms. **[Student Name**] and Mr./Ms. [**Student Name**] in partial fulfillment of the requirements for the degree of Master of Computer Application, Semester [**Semester**], at D.K.T.E. Society's Textile and Engineering Institute, Ichalkaranji.

The project work was carried out under the guidance of Mr./Ms. [Project Guide Name] and is submitted in accordance with the academic requirements of the institute for the academic year **2026–2027**.

The report has been reviewed for academic submission. The student details, project title, semester, and academic year should be completed and verified before the certificate is signed.

Project Guide	Head of the Department
Name: _____	**Name: _____**
Signature: _____	**Signature: _____**
Date: _____	**Date: _____**
Examiner 1: _____	**Examiner 2: _____**

Place: Ichalkaranji

GUIDE'S CERTIFICATE

I hereby certify that the project entitled "[PROJECT TITLE]" has been developed by [Student Name(s)] under my guidance.

The student(s) have undertaken the project work, studied the relevant concepts of Generative Artificial Intelligence, implemented the proposed solution, and prepared the accompanying documentation.

The work has been reviewed by me and is submitted for academic evaluation. The project has been carried out under my supervision, subject to the institute's applicable academic integrity requirements.

Project Guide

Name: _____

Designation: _____

Signature: _____

Date: _____

STUDENT DECLARATION

To,
The Registrar,
Shivaji University, Kolhapur.

We, the undersigned, hereby declare that the project report entitled "[**PROJECT TITLE**]", submitted under the guidance of Mr./Ms. [**Guide Name**] is our original work carried out as part of the Master of Computer Application programme at D.K.T.E. Society's Textile and Engineering Institute, Ichalkaranji.

We further declare that the empirical findings and results presented in this report are based on the work and information collected by us. We have not copied this report from another report submitted to Shivaji University, Kolhapur. All external sources, software libraries, datasets, APIs, and Generative AI tools used have been acknowledged. Any AI-generated content or code has been reviewed and verified by us, in accordance with applicable academic-integrity requirements.

We understand that plagiarism or misrepresentation may be dealt with under the applicable university and institute rules.

1. The project has been developed by us under the guidance of the assigned project guide.
2. The information, results, screenshots, and observations presented in this report are accurate to the best of our knowledge.
3. All external sources, research papers, software libraries, datasets, APIs, and AI tools used in the project have been appropriately acknowledged.
4. The use of Generative AI tools, including AI-generated code, content, images, or other outputs, has been disclosed wherever applicable.
5. We have not knowingly presented another person's work as our own.
6. This project has not been submitted elsewhere for academic evaluation unless explicitly permitted by the institute.

Student Name(s): _____

Roll Number(s): _____

Signature(s): _____

Date: _____

ACKNOWLEDGEMENT

We have been fortunate to receive assistance, suggestions, and support from our project guide, faculty members, friends, and the institute during the development of our project entitled "[**PROJECT TITLE**]".

First and foremost, we express our sincere gratitude to our respected project guide, Mr./Ms. [**Guide Name**], for valuable guidance, constructive suggestions, and kind support throughout the project work.

We take this opportunity to thank the Head of the Department, all teaching and non-teaching staff members, and our friends for their cooperation and encouragement.

We also sincerely thank the staff of the computer laboratory for their cooperation and for providing the facilities required to complete the project within the available time.

We acknowledge the developers and providers of software tools, Generative AI models, libraries, APIs, datasets, and technical resources used in the project, wherever applicable.

Mr./Ms. [Student Name(s)]

Mr./Ms. [Student Name(s)]

ABSTRACT

Recommended length: 200–300 words. Replace the sample text below with a concise description of your actual project.

This project focuses on the development of [**brief description of the application**] using Generative Artificial Intelligence.

The existing approach to **[problem area]** faces challenges such as [**identify major challenges].** To address these issues, the proposed system uses [**AI model or technique**] to provide [**main functionality**].

The application is developed using [programming language, framework, database, and other technologies]. The Generative AI component is used for [**specific task, such as text generation, question answering, summarization, code assistance, or image generation**].

The system provides features such as [list the major features]. The project is evaluated through functional testing and relevant performance measures.

The proposed application aims to improve [**efficiency, accessibility, personalization, productivity, or another relevant objective**]. The project also identifies limitations and opportunities for future enhancement.

TABLE OF CONTENTS

Sr. No.	Contents	Page No.
1	Chapter 1: Introduction	11
2	Chapter 2: Existing System and Problem Analysis	12
3	Chapter 3: Proposed System	13
4	Chapter 4: Generative AI Components Used	15
5	Chapter 5: System Analysis and Design	17
6	Chapter 6: Hardware and Software Requirements	18
7	Chapter 7: Implementation	19
8	Chapter 8: Testing and Results	20
9	Chapter 9: Scope, Limitations and Future Enhancements	21
10	Chapter 10: Conclusion	22
11	Bibliography / References	23
12	Appendices	24

Update the page-number column after completing the report. In Microsoft Word, an automatic Table of Contents can be inserted from References > Table of Contents

LIST OF FIGURES

Figure No.	Figure Title	Page No.
Fig. 5.1	System Architecture	17
Fig. 5.2	Use Case Diagram	17
Fig. 5.3	Data Flow Diagram	17
Fig. 5.4	Activity / Sequence Diagram	17
Fig. 7.1	Application Home Screen	19
Fig. 7.2	AI Output Screen	19
Fig. 8.1	Testing Results	20

Keep only figures that actually appear in the final report.

LIST OF TABLES

Table No.	Table Title	Page No.
Table 2.1	Existing System Analysis	12
Table 3.1	Proposed System Features	13
Table 4.1	Generative AI Components	15
Table 6.1	Hardware Requirements	18
Table 6.2	Software Requirements	18
Table 8.1	Test Cases and Results	20

ABBREVIATIONS AND TERMINOLOGY

Abbreviation	Full Form / Meaning
AI	Artificial Intelligence
GenAI	Generative Artificial Intelligence
LLM	Large Language Model
NLP	Natural Language Processing
API	Application Programming Interface
RAG	Retrieval-Augmented Generation
UI	User Interface
UX	User Experience
DBMS	Database Management System
SQL	Structured Query Language
JSON	JavaScript Object Notation
GPU	Graphics Processing Unit

Add only terms relevant to your project.

CHAPTER 1: INTRODUCTION

1.1 Introduction to the Project

Describe the project domain, the problem area, and why the project is being developed.

1.2 Background of the Project

Describe the existing situation, relevant technologies, and developments in Generative AI that make the project feasible.

1.3 Motivation

Explain why the project was selected. Possible motivations include automation of repetitive tasks, improved productivity, intelligent assistance, personalized learning, or better information accessibility.

1.4 Problem Statement

Suggested format: "To design and develop a [type of application] using Generative Artificial Intelligence to address [specific problem] by providing [key functionality] to [target users]."

1.5 Objectives of the Project

7. To study existing methods used to solve the selected problem.
8. To design a user-friendly application.
9. To integrate a suitable Generative AI model or service.
10. To implement the major functional modules.
11. To evaluate the usefulness and performance of the application using appropriate tests.
12. To identify system limitations and opportunities for improvement.

1.6 Scope of the Project

Define the intended users, application domain, main features, AI capabilities, supported platforms, and boundaries of the implementation.

1.7 Organization of the Report

Briefly describe the contents of each chapter.

CHAPTER 2: EXISTING SYSTEM AND PROBLEM ANALYSIS

2.1 Overview of the Existing System

Describe the current method, manual process, software, or application used to address the problem.

2.2 Working of the Existing System

Explain the process step by step. Add a flowchart if appropriate.

2.3 Limitations of the Existing System

Discuss actual limitations such as manual effort, time-consuming operations, limited personalization, restricted automation, difficulty handling unstructured information, or lack of intelligent interaction. Do not list limitations that do not apply.

2.4 Literature Review

Review relevant research papers, technical articles, existing applications, and official technical documentation. For each source, record the author/organization, year, title, contribution, and relevance or limitation.

2.5 Comparative Analysis

Feature	Existing System	Proposed System
User Interaction	Describe current approach	Describe new approach
Automation	Current capability	Proposed capability
AI Integration	Existing capability, if any	Proposed GenAI integration
Personalization	Existing capability	Proposed capability
Response Generation	Existing method	Proposed method
Data Management	Existing method	Proposed method
Limitations	Identified issues	Remaining limitations

2.6 Problem Identification

Summarize the major problems identified during the analysis.

2.7 Need for the Proposed System

Explain why a new solution is justified and what improvement it is expected to provide.

CHAPTER 3: PROPOSED SYSTEM

3.1 Introduction

Describe the solution proposed by the project team.

3.2 Overview of the Proposed System

Explain how the application works and how Generative AI contributes to the solution.

3.3 Proposed System Architecture

Include a labelled architecture diagram. A typical flow may be: User → User Interface → Application Backend → AI Model/API → Response Processing → Output to User. Add database, retrieval, or external service components if used.

3.4 Major Features

Describe each implemented feature, such as user interaction, AI-generated responses, summarization, document processing, data storage, user feedback, history management, or report generation.

3.5 Functional Requirements

- The system shall accept user input.
- The system shall validate required fields.
- The system shall process requests through the configured AI component.
- The system shall display generated output.
- The system shall handle API errors and invalid requests.
- The system shall save relevant information when required.

3.6 Non-Functional Requirements

Describe usability, reliability, performance, security, maintainability, scalability, accessibility, and privacy requirements.

3.7 Module Description

Module	Description
User Interface	Accepts input and displays results.
Input Processing	Validates and prepares user input.
Prompt Management	Constructs instructions for the AI model.
AI Integration	Sends requests to the selected model or API.

Output Processing	Formats and validates generated responses.
Database Module	Stores required application data, if applicable.
Error Handling	Handles invalid inputs, failures, and timeouts.

3.8 Advantages of the Proposed System

Explain expected advantages over the existing system.

3.9 Feasibility Study

Technical feasibility: availability of technology and skills. Economic feasibility: development, API, hosting, and maintenance costs. Operational feasibility: suitability for intended users. Schedule feasibility: completion within available time.

CHAPTER 4: GENERATIVE ARTIFICIAL INTELLIGENCE COMPONENTS USED

This chapter is essential for a Generative AI project. Explain exactly how Generative AI is used, rather than merely mentioning an AI tool.

4.1 Introduction to Generative AI

Explain Generative AI and its relevance to the selected application.

4.2 Generative AI Model Used

Identify the actual model or service, provider, version/model identifier where available, purpose, and access method. Examples include GPT-family models, Claude models, Gemini models, open-source language models, image-generation models, or locally deployed language models.

4.3 Generative AI Components

Component	Description	Use in Project
Large Language Model	Generates or transforms text	Specify actual use
Prompt Engineering	Structures instructions for the model	Explain prompt design
API Integration	Connects application to AI service	Explain integration
RAG	Grounds responses in retrieved information	Only if implemented
Embeddings	Represent content as vectors	Only if used
Vector Database	Stores/retrieves vector representations	Only if used
AI Agents	Select or execute tasks using tools	Only if implemented
Image Generation	Generates images from prompts	Only if implemented
Speech Processing	Speech-to-text or text-to-speech	Only if implemented

4.4 Prompt Engineering and Design

Explain how prompts are designed. Consider task/role definition, context, input instructions, output format, constraints, and error handling.

Example basic prompt: "Explain database normalization."

Example improved prompt: "Act as a database management lecturer for MCA students. Explain database normalization in simple English. Cover 1NF, 2NF, and 3NF using one common example. Present the explanation with headings, a comparison table, and five revision questions."

Include actual prompts used in your application and explain their purpose.

4.5 Workflow of the Generative AI Component

13. User submits a request.
14. The application validates the input.
15. The application constructs the prompt.
16. The request is sent to the AI model or service.
17. The model generates an output.
18. The application processes and displays the response.
19. The user reviews the result.
20. Relevant feedback or output is stored, if applicable.

4.6 Model Parameters and Configuration

Where applicable, document model identifier, temperature, maximum output tokens, system instructions, context-window constraints, and other model-specific parameters. Do not include parameters that are not configured or cannot be verified.

4.7 AI Output Validation

Explain how the application handles incorrect or irrelevant responses, hallucinations, inappropriate input/output, API failures, missing information, and the need for user verification.

4.8 Responsible AI Considerations

Discuss data privacy, bias and fairness, copyright and attribution, human oversight, misinformation, API key protection, and ethical use.

4.9 Role of Generative AI in the Project

Conclude by explaining which tasks genuinely depend on Generative AI and which are performed using conventional programming.

CHAPTER 5: SYSTEM ANALYSIS AND DESIGN

5.1 System Architecture Diagram

Include a labelled architecture diagram showing major components and their interactions.

5.2 Use Case Diagram

Show actors and the functions they can perform.

5.3 Data Flow Diagram (DFD)

Include relevant DFD levels, such as Level 0 and Level 1, where appropriate.

5.4 Flowchart

Show the main processing sequence.

5.5 Database Design

If a database is used, provide the ER diagram, table structures, primary/foreign keys, relationships, and data dictionary. Otherwise, explain how application data is managed.

5.6 Input and Output Design

Include screenshots or mock-ups showing input and output interfaces.

5.7 Security Design

Explain protection of user data, API keys, authentication, and other sensitive information.

CHAPTER 6: HARDWARE AND SOFTWARE REQUIREMENTS

6.1 Hardware Requirements

Component	Minimum / Expected Requirement	Actual Configuration
Processor	Suitable for selected application	Specify
RAM	Sufficient to run application	Specify
Storage	Sufficient free space	Specify
Display	Standard supported display	Specify
Internet	Required for cloud AI services	Specify
GPU	Only for certain local AI workloads	Specify / N.A.

6.2 Software Requirements

Software Component	Purpose	Version Used
Operating System	Runs the application	Specify
Programming Language	Application development	Specify
IDE / Code Editor	Writing and debugging code	Specify
Front-End Technology	User interface	Specify
Back-End Framework	Application logic	Specify
Database	Data storage	Specify / N.A.
AI Model / API	Generative AI functionality	Specify
Libraries / Packages	Additional functionality	Specify
Web Browser	Application access and testing	Specify
Version Control	Source-code management	Specify / N.A.

6.3 Development Environment

Describe the development tools, framework, package manager, and deployment environment.

6.4 AI Service Requirements

For a cloud AI service, document provider and model, API access requirements, internet connectivity, usage limits, and pricing considerations where relevant. Never include secret API keys, passwords, or authentication tokens.

CHAPTER 7: IMPLEMENTATION

7.1 Introduction

Explain how the proposed system was converted into a working application.

7.2 Project Structure

Show the folder and file structure.

7.3 Implementation of Modules

Explain the implementation of each major module.

7.4 AI Model Integration

Describe how the application connects to the model or API. Include simplified code examples where necessary.

7.5 Prompt Implementation

Provide selected prompts and explain how they are constructed and used.

7.6 Database Implementation

Explain tables, queries, and data operations, if applicable.

7.7 User Interface Implementation

Include screenshots of the actual application with figure numbers and captions.

7.8 Error Handling

Explain handling of invalid input, network errors, model failures, and unexpected output.

7.9 Deployment

Describe how to install, configure, and run the application.

7.10 Source Code Repository

Provide the GitHub repository link, if applicable. Ensure no API keys, passwords, private datasets, or sensitive information are exposed.

CHAPTER 8: TESTING AND RESULTS

8.1 Introduction to Testing

Explain the purpose and scope of testing.

8.2 Testing Methodology

Describe applicable tests, such as unit, integration, functional, UI, performance, AI output evaluation, and security testing.

8.3 Test Cases

Test ID	Scenario	Input	Expected Output	Actual Output / Status
TC01	Valid user input	Valid request	Appropriate response	Record result
TC02	Empty input	Blank request	Validation message	Record result
TC03	AI service unavailable	Valid request	Error/fallback message	Record result
TC04	Long input	Large request	Appropriate handling	Record result
TC05	Irrelevant query	Unrelated question	Suitable response or limitation	Record result

8.4 AI Output Evaluation

Describe how output quality is evaluated. Possible criteria include relevance, factual correctness, completeness, readability, consistency, response time, and user satisfaction. Use a defined test dataset and scoring rubric where possible.

8.5 Results and Observations

Present actual results obtained during testing. Include tables, charts, or screenshots where appropriate.

8.6 Performance Analysis

Measure relevant parameters, such as response time, resource consumption, or success rate. Do not report invented accuracy percentages or performance improvements.

8.7 Screenshots of the Working Application

Include labelled screenshots of the home page, input interface, AI-generated output, error message, database records if applicable, and final result.

CHAPTER 9: SCOPE, LIMITATIONS AND FUTURE ENHANCEMENTS

9.1 Scope of the Project

Describe the intended application area and the features currently supported.

9.2 Limitations of the Project

Discuss actual limitations, such as dependence on third-party AI services, internet requirements, API usage limits or costs, inaccurate or hallucinated responses, limited data, language/domain restrictions, scalability, or privacy constraints.

9.3 Future Enhancements

- Integration of additional AI models.
- Development of a mobile application.
- Multilingual support.
- Voice-based interaction.
- RAG using verified documents.
- Improved output evaluation and validation.
- User authentication and role-based access.
- Integration with external databases and services.
- AI agents for controlled task automation.
- Improved privacy and security mechanisms.

9.4 Social and Practical Applications

Explain how the project may benefit its intended users or application domain.

9.5 Sustainability and Maintainability

Discuss how the application can be maintained, updated, and operated over time.

CHAPTER 10: CONCLUSION

10.1 Conclusion

Summarize the problem addressed, proposed solution, Generative AI technologies used, major features implemented, results obtained, objectives achieved, and major limitations.

10.2 Learning Outcomes

Explain the technical and practical knowledge gained during development.

- Understanding Generative AI concepts.
- Applying prompt engineering.
- Integrating an AI model or API.
- Developing a complete application.
- Evaluating AI-generated outputs.
- Understanding responsible AI practices.
- Working collaboratively on a software project.

10.3 Final Remarks

Conclude with a brief statement about the project's usefulness and possible future development.

BIBLIOGRAPHY / REFERENCES

List all sources consulted during the project, including research papers, books, official AI model documentation, API documentation, software documentation, dataset documentation, reputable technical articles, and official project repositories.

Recommended reference formats

Research paper: Author(s), "Title of the Paper," Journal or Conference Name, volume, issue, page numbers, year, DOI or other persistent identifier.

Book: Author(s), Title of the Book, edition, publisher, year.

Website or official documentation: Organization or author, "Title of the Page or Documentation," publication/update date if available, URL, accessed date.

Software library or model: Developer or organization, model/library name, version or identifier, official documentation or repository, accessed date where relevant.

Reference list

[1] [Author / Organization]. "[Title of source]." [Publisher / Journal / Official documentation]. [Year]. [DOI or URL]. [Accessed date, if applicable].

[2] [Author / Organization]. "[Title of source]." [Publisher / Journal / Official documentation]. [Year]. [DOI or URL]. [Accessed date, if applicable].

[3] [Author / Organization]. "[Title of source]." [Publisher / Journal / Official documentation]. [Year]. [DOI or URL]. [Accessed date, if applicable].

[4] [Author / Organization]. "[Title of source]." [Publisher / Journal / Official documentation]. [Year]. [DOI or URL]. [Accessed date, if applicable].

[5] [Author / Organization]. "[Title of source]." [Publisher / Journal / Official documentation]. [Year]. [DOI or URL]. [Accessed date, if applicable].

[6] [Author / Organization]. "[Title of source]." [Publisher / Journal / Official documentation]. [Year]. [DOI or URL]. [Accessed date, if applicable].

Use one consistent referencing style. Include only sources actually consulted. Do not fabricate papers, authors, DOIs, or URLs. A sample reference format is not a substitute for a real citation.

APPENDICES

Appendix A: Source Code

Include important code snippets or provide the repository reference. Keep credentials and private information out of the report.

Appendix B: Prompt Library

Include important prompts used in the project, along with their purpose, inputs, and representative outputs.

Appendix C: Sample Input and Output

Provide representative examples of user inputs and generated outputs. Identify any output that has been edited or post-processed.

Appendix D: User Manual

Explain how to install, configure, and use the application.

Appendix E: Installation and Deployment Guide

Provide setup instructions, dependencies, configuration steps, and execution commands.

Appendix F: Test Results

Include detailed test cases and supporting evidence.

Appendix G: Dataset Description

Describe dataset source, format, size, preprocessing, permissions, and relevant privacy considerations.

Appendix H: Additional Screenshots

Include screenshots that support the explanation but are not essential to the main chapters.

Appendix I: AI Usage Disclosure

Suggested wording: "Generative AI tools were used during the development of this project for [specify tasks]. The project team reviewed, tested, and verified the relevant outputs. The final implementation, evaluation, and

documentation were checked by the team, and external sources have been acknowledged as applicable."

Modify this statement to reflect the actual use of AI tools and the institute's academic-integrity policy.

APPENDIX J: RECOMMENDED FORMATTING GUIDELINES

Parameter	Recommended Format
Paper Size	A4
Font	Times New Roman or another approved academic font
Main / Chapter Heading	16–17 pt, bold
Section Heading	13 pt, bold
Body Text	11–12 pt
Line Spacing	1.5, if required by the department
Alignment	Justified for body text
Left Margin	1.5 inches, if prescribed
Right / Top / Bottom Margin	1 inch, if prescribed
Preliminary Page Numbering	Roman numerals, if prescribed
Main Chapter Numbering	Arabic numerals
Figure Captions	Below figures
Table Captions	Above tables
Code Formatting	Monospaced font with readable indentation
References	One consistent academic style

Start each main chapter on a new page. The institute's prescribed format takes precedence over these recommendations.

APPENDIX K: FINAL SUBMISSION CHECKLIST

A. Documentation

Title page contains all required details.

Certificates are completed and signed where required.

Student declaration is included.

Abstract is complete.

Table of contents matches final page numbers.

Figures and tables are numbered correctly.

All chapters are arranged in the prescribed order.

B. Generative AI Implementation

The AI model or service is identified.

The role of Generative AI is clearly explained.

Actual prompts are documented.

System architecture includes the AI component.

AI-generated outputs have been tested.

Limitations and responsible AI considerations are documented.

External sources and AI tools are acknowledged.

C. Technical Implementation

Hardware requirements are specified.

Software requirements and versions are documented.

Major modules are explained.

Screenshots show the working application.

Test cases and actual results are included.

Source code or repository details are provided, where applicable.

Setup and execution instructions are included.

D. Academic Integrity

External content is properly cited.

No fabricated test results are included.

AI-generated material has been reviewed and verified.

The student can explain the implementation.

No passwords, API keys, or confidential information appear in the report.

E. Final Submission

The report follows the prescribed formatting.

Page numbers and headings are consistent.

The final report is proofread.

Required digital files are submitted.

The working application and demonstration are ready.

DEPARTMENTAL RECOMMENDATION

For Generative AI projects, the following four components are recommended as compulsory additions to conventional software project documentation:

- Generative AI Component Analysis: explain the model, prompting strategy, API integration, and actual role of AI.
- Prompt Engineering Evidence: document representative prompts, input/output examples, and prompt refinements.
- AI Evaluation and Responsible AI: demonstrate output-quality testing and address hallucinations, privacy, and relevant risks.
- Working Application Evidence: submit screenshots, test results, repository details where applicable, and instructions to run the project.

Recommended final submission: one properly formatted PDF report, the working application, source code or repository link, a prompt documentation appendix, and supporting test evidence.

This is a recommended departmental template. Institute-prescribed project-report formats, examination rules, and academic-integrity policies take precedence.